Leonardo Pacciani-Mori

Personal Data

Address: Department of Physics, *University of California, San Diego*
Natural Sciences Building, Room 2121
9500 Gilman Dr, La Jolla, CA 92093
Personal website: leonardo.pm
email: lpaccianimori@physics.ucsd.edu

Research experience

Mar 1st 2021 – Present day

Postdoctoral researcher
University of California, San Diego, La Jolla (CA)
Department of Physics
Supervisor: Prof. Terence Hwa

Apr 1st 2019 – Sep 30th 2019

Fellow of the Department of Physics
Harvard University, Cambridge (MA)
Department of Physics
Supervisors: Dr. Andrea Giometto, Prof. Andrew Murray, Prof. David Nelson
I have worked as a visiting graduate student in the groups of Andrew Murray and David Nelson, supervised by Andrea Giometto

Education

Oct 1st 2017 – Dec 16th 2020

Ph.D. in *Physics*, **University of Padua**, Padua (Italy)
Thesis: *Two improved approaches to the study of competition in microbial communities*
Supervisor: Prof. Amos Maritan
Co-supervisor: Prof. Samir Suweis
Tutor: Prof. Flavio Seno
Final evaluation: Excellent

Oct 1st 2015 – Sep 26th 2017

Master's Degree in *Physics*, **University of Padua**, Padua (Italy)
Theoretical course of study
Thesis: *A physics approach to ecosystem dynamics*
Supervisor: Prof. Amos Maritan
Final grade: 110/110 *cum laude*

Oct 1st 2012 – Jul 22nd 2015

Bachelor's Degree in *Physics*, **University of Padua**, Padua (Italy)
Thesis: *Curvature effect on patterning dynamics on spherical membranes*
Supervisor: Prof. Enzo Orlandini
Final grade: 108/110

Publications

- **Pacciani-Mori L.**, Suweis S., Maritan A. and Giometto A. (2021) Constrained proteome allocation affects coexistence in models of competitive microbial communities, *The ISME Journal*. DOI: 10.1038/s41396-020-00863-0
- **Pacciani-Mori L.**, Giometto A., Suweis S. and Maritan A. (2020) Dynamic metabolic adaptation can promote species coexistence in competitive microbial communities, *PLOS Computational Biology*, 16(5): e1007896. DOI: 10.1371/journal.pcbi.1007896.

Attended conferences, workshops and schools

Jun 22nd – 25th 2021	**Stochastic Models and Experiments in Ecology and Biology** European Center for Living Technology, *Ca' Foscari University of Venice* Hybrid online/in person conference Website
Jan 14th – Mar 18th 2021	**Forum on Microbial Metabolism** Online workshop, Kavli Institute of Theoretical Physics at *University of California, Santa Barbara* Website
Jan 19th – 21st 2021	**Workshop on Limits to Diversity Assembly** Online workshop, *International Centre for Theoretical Physics*, Trieste Website
Nov 30th – Dec 18th 2020	**Winter School on Quantitative Systems Biology: Quantitative Approaches in Ecosystem Ecology** Online school, *International Centre for Theoretical Physics*, Trieste **Lectures held:** Tutorial on nonlinear dynamics Website
Nov 12th – 13th 2020	**Paris Biological Physics Community Day** Online seminar, *ENS Paris, Institut Curie, and Université Paris-Diderot*, Paris **Contributed talk:** Constrained proteome allocation affects coexistence in models of competitive microbial communities Website
Jan 7th – 10th 2020	**English for Academic Writing** Forlì campus, *University of Bologna* Website
Mar 6th – 10th 2019	**Quinta Conferenza Italiana degli Studenti di Fisica (5th Italian Conference of Physics Students)** Department of Physics, *University of Milan* **Contributed talk:** Strategie metaboliche adattive: una risposta (apparentemente) semplice ed efficace a molti problemi in ecologia e microbiologia *(Adaptive metabolic strategies: an (apparently) simple and effective answer to many challenging problems in ecology and microbiology)* **Contributed poster:** Adaptive consumer-resource models can explain diauxic shifts and the violation of the Competitive Exclusion Principle Website
Dec 20th 2018	**The physics of complex systems IV: from Padova to the rest of the world and back** Department of Physics and Astronomy, *University of Padua* **Invited talk:** Adaptive metabolic strategies: an (apparently) simple and effective answer to many challenging problems in ecology and microbiology
Sep 23rd – 28th 2018	**Conference on Complex Systems 2018** Vellidio Convention Center, *Aristotle University of Thessaloniki* **Contributed talk:** Adaptive metabolic strategies explain diauxic shifts and promote species coexistence Website
Sep 17th – 21st 2018	**104° Congresso della Società Italiana di Fisica (104th Congress of the Italian Physical Society)** Department of Physics, *University of Calabria* **Invited talk:** Fisica ed ecosistemi: il dilemma del rapporto stabilità-biodiversità *(Physics and ecosystems: the dilemma of the stability-biodiversity relationship)* Website

Apr 20th – 23rd 2018	**Quarta Conferenza Italiana degli Studenti di Fisica** **(4th Italian Conference of Physics Students)** Department of Physics, *University of Pisa* **Contributed talk:** Fisica ed ecosistemi: il dilemma del rapporto stabilità-biodiversità *(Physics and ecosystems: the dilemma of the stability-biodiversity relationship)* Website
Apr 5th – 7th 2018	**Stochastic Models in Ecology and Evolutionary Biology** *Venetian Institute of Sciences, Letters and Arts*, Venice **Contributed talk:** The role of metabolic trade-offs in the establishment of biodiversity
Feb 19th – Mar 16th 2018	**Spring College on the Physics of Complex Systems** *International Centre for Theoretical Physics*, Trieste Website
Aug 7th – 13th 2017	**32nd International Conference of Physics Students** *University of Turin* Attended as *national delegate* of the Italian Association of Physics Students Website
May 11th – 13th 2017	**Terza Conferenza Italiana degli Studenti di Fisica** **(3rd Italian Conference of Physics Students)** Department of Physics, *University of Bari* Attended as *spokesperson of the Local Committee of Padua of the Italian Association of Physics Students* Website
Oct 3rd – 6th 2016	**Venice meeting on fluctuations in small complex systems III** *Venetian Institute of Sciences, Letters and Arts*, Venice
Apr 22nd – 24th 2016	**Seconda Conferenza Italiana degli Studenti di Fisica** **(2nd Italian Conference of Physics Students)** Department of Physics, *Univeristy of Turin* Website

Funding

Mar 1st 2022 – Feb 25th 2025	**Simons Foundation** **Total amount awarded:** $ 270,000.00 *Simons Postdoctoral Fellowship in Marine Microbial Ecology* Website
Oct 1st 2017 – Sep 30th 2020	**Fondazione Cariparo** **Total amount awarded:** € 74,768.88 *Fellowship awarded each year to the candidate who ranks first in the selection process of the PhD School in Physics*

Journals for which I've acted as reviewer

PLoS Computational Biololgy, Nature Communications, Physical Review E.

Honors and awards

Best contributed talk at the 4th Italian Conference of Physics Students.

Best poster at the 5th Italian Conference of Physics Students.

Organized conferences, workshops and schools

Jun 22nd – 25th 2021 **Stochastic Models and Experiments in Ecology and Biology**
European Center for Living Technology, *Ca' Foscari University of Venice*
Hybrid online/in person conference
Website

Oct 2016 – May 2017 **Sei Spritz Facili**
Department of Physics and Astronomy, *University of Padua*
Series of six informal lectures held by professors for the students of Physics and Astronomy, covering the areas over which research is carried out in the Department.
Activity of the Padua Local Committee of the Italian Association of Physics Students.

Teaching experience

Nov 30th – Dec 1st 2020 **Tutorial on nonlinear dynamics**
International Centre for Theoretical Physics, Trieste
Introductory online lectures on nonlinear dynamics, part of the Winter School on Quantitative Systems Biology
Website and recordings

Dec 6th – 13th 2018 **LaTeX for dummies: guida di sopravvivenza per fisici**
(LaTeX for dummies: a survival guide for physicists)
Department of Physics, *University of Padua*
Introductory 4-lecture course on LaTeX for complete beginners, organized with the Padua Local Committee of the Italian Association of Physics Students.
Website

Service and memberships

Oct 22nd 2019 – Present day **Member of the Association of PhD students and graduates in Italy**

Oct 20th 2018 – Sep 30th 2019 **Vicepresident of the Italian Association of Physics Students**

Sep 23rd 2018 – Present day **Member of the Complex Systems Society**

Oct 1st 2017 – Sep 30th 2019 **Secretary of the Italian Association of Physics Students**
Elected during the 3rd Italian Conference of Physics Students

May 2nd 2016 – Sep 30th 2017 **President of the Local Committee of Padua of the Italian Association of Physics Students**

Feb 26th 2016 – Sep 30th 2020 **Member of the Italian Association of Physics Students**

Language skills

Italian: Native
English: Fluent
Spanish: Intermediate

Other languages studied (mainly for personal interest and at basic or elementary level):
Japanese, Chinese (Mandarin), Esperanto, Greek (modern), German.